Experiment No: 6			
Course Name:	Web Programming Laboratory (216U01L405)	**Semester:**	**IV**
Date of Performance:	**07/04/26**	**DIV/ Batch No:**	**B2**
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Title: Develop a PHP application that performs CRUD (Create, Read, Update, Delete) operations.

Aim of the Experiment:

To design and develop a dynamic web application using PHP and MySQL that performs CRUD (Create, Read, Update, Delete) operations on a database.

Objectives for the Experiment:

- Understand and explain the architecture of a PHP–MySQL web application and the role of CRUD operations in database-driven systems.
- Implement Create, Read, Update, and Delete functionalities using PHP and MySQL with proper form handling and database connectivity.
- Analyze and validate user inputs and database transactions to ensure data integrity and error handling (if possible secure from SQL injection).
- Design and develop a functional and user-friendly CRUD-based web application following best practices in modular coding and structured programming.

COs to be achieved:

1. Design dynamic web pages using various HTML tags.
2. Use CSS to prepare the layout of web pages.
3. Apply JavaScript for validation in client-side programming.
4. Integrate server-side pages using PHP.
5. Apply database operations by integrating SQL queries and session variables.

Books/ Journals/ Websites references:

1. Students should write

Software / Tools Required

- XAMPP / WAMP / MAMP
- PHP
- MySQL
- phpMyAdmin
- Code Editor (VS Code / Sublime / Notepad++/ any other)
- Web Browser

Methodology:

1. Start XAMPP server.
2. Create database and table in phpMyAdmin.
3. Establish PHP-MySQL connection.
4. Create HTML form for data input.
5. Write PHP code for:
    - Insert
    - Fetch
    - Update
    - Delete
6. Execute queries.
7. Display results.
8. Close database connection.

Theory:

Inventory management systems track products, quantity, and price. PHP is used as server-side language to process user input and interact with MySQL database. CRUD operations allow adding new items, viewing stock, updating details, and deleting records. This ensures efficient data management in real-world applications.

Problem statement:

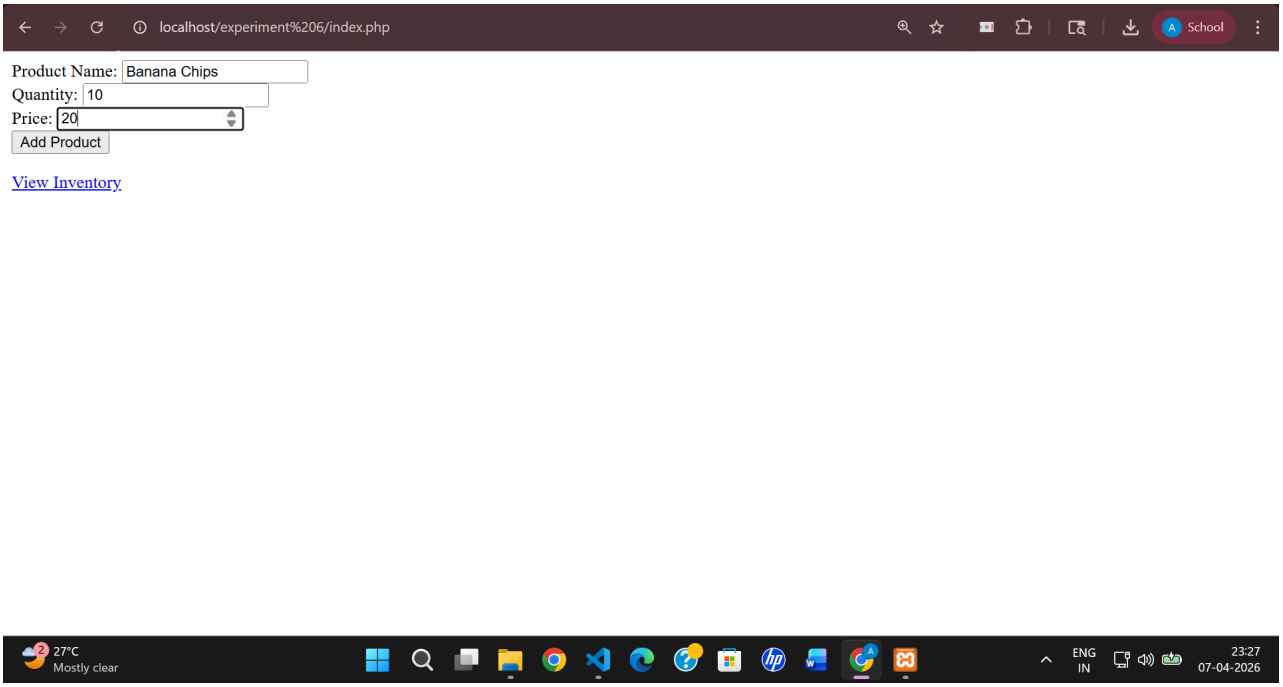
Design and implement an application to demonstrate HTML form integration with PHP for data collection and processing and perform the CRUD operations.

Note: Mini Project should start with CRUD operations too.

Note: Ensure all mini project code / work is regularly pushed to Github

Code:

Output / Screenshot:



ID	Name	Quantity	Price
1	Banana Chips	10	20

Product added successfully

Post Lab Subjective/Objective type Questions:

1. What is the role of mysqli or PDO in PHP?
   mysqli/PDO are used in PHP to connect applications with databases. They help execute SQL queries for inserting, reading, updating, and deleting data securely and efficiently.
2. What is the difference between GET and POST methods in PHP?
   GET method sends data through the URL and is mainly used for retrieving data, while POST sends data in the request body and is more secure for form submissions.

3. Explain the flow of data when a user submits a form to insert a record into the database.
   When a user submits a form, data is sent to the server. PHP processes it using $_POST, executes an SQL INSERT query, stores data in database, and displays the result.

4. Write a PHP code snippet to establish a database connection using mysqli.

```
<?php
$conn = mysqli_connect("localhost","root","","inventory_db");

if(!$conn){
   die("Connection failed");
}
?>
```

5. Modify the Update functionality to allow partial record updates instead of updating all fields.
   Partial update allows modifying only selected fields. The SQL query is dynamically built to update only non-empty inputs, ensuring unchanged fields remain intact in the database.

6. Propose enhancements to extend your CRUD application (e.g., authentication, pagination, search functionality, REST API integration).

   The CRUD application can be enhanced by adding authentication, search functionality, pagination, and input validation. Integration with REST APIs can further improve scalability and usability of the system.

Conclusion and Discussion:

The experiment successfully implemented an inventory management system using PHP and MySQL. CRUD operations were performed to manage product data efficiently. The system demonstrates real-world application of database-driven web development.

Aarav: Application works correctly, UI can improve.
Arnav: CRUD operations are clear and easy to understand.
Sakshi: Good implementation, can add search feature